

CIS 580

# Machine Perception

*Or Geometric Computer Vision*

Instructor: Lingjie Liu

Lec 1: Jan 20, 2026

# Learning Outcomes

- Firm knowledge of fundamentals of geometric computer vision from single and multiple cameras, some image processing, and (if time permits) some deep learning for geometry
- Understanding of challenges: why algorithms work or do not work
- Ability to perform as a vision engineer in vision and robotics companies

**Prerequisites:** mainly undergraduate linear algebra (e.g. vector & matrix products, inverses, determinants, solving systems of linear equations, eigenvectors), and some high school Euclidean geometry

# CIS 580 Machine Perception Spring 2025

- Resources:
  - Canvas
    - Slides & readings
    - (-> Recordings, Ed, Gradescope)

# Instructor Introduction

Lingjie Liu

Assistant Professor, CIS

Penn CG lab and GRASP lab

<https://lingjie0206.github.io/>



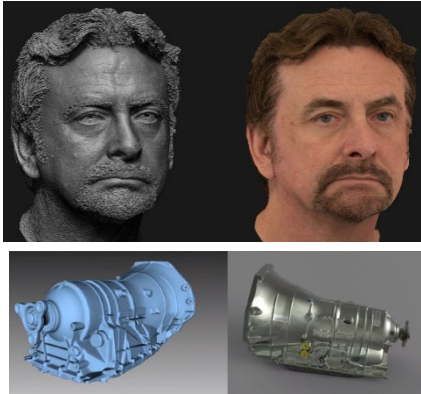
# My Research Interests

- **Reconstruction** of Real-world Dynamic Scenes.

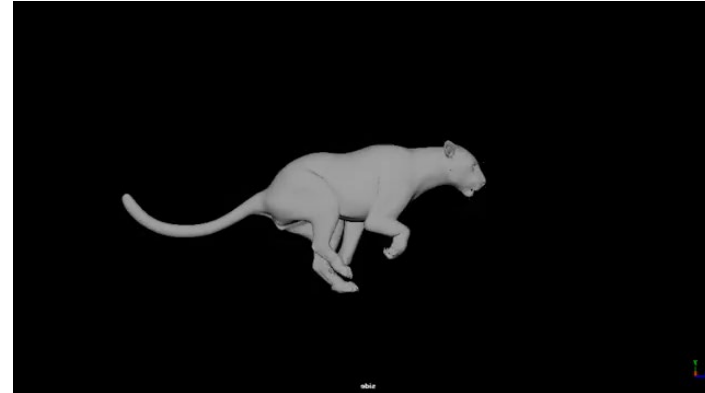


# My Research Interests

- **Reconstruction** of Real-world Dynamic Scenes.



Geometry  
+ Appearance



Motion  
+ Deformation



# My Research Interests

- **Reconstruction of Real-world Dynamic Scenes.**





# My Research Interests

- **Image Synthesis** of Real-world Scenes with 3D Control.





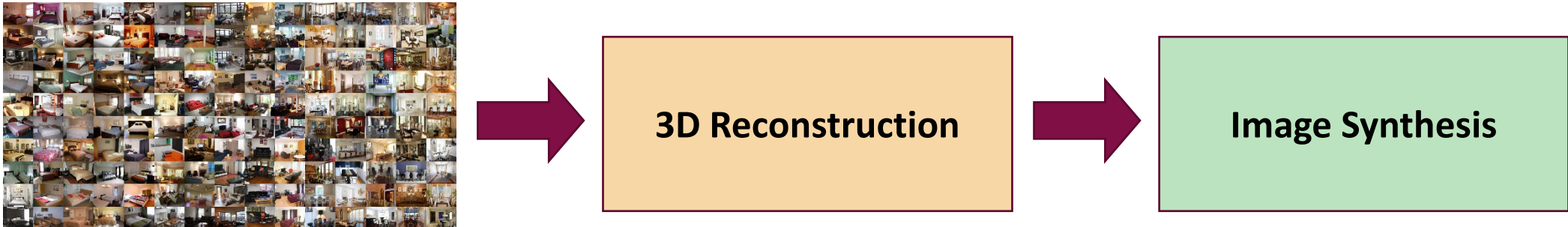
# My Research Interests

- **Image Synthesis** of Real-world Scenes with 3D Control.



# My Research Interests

- **Image Synthesis** of Real-world Scenes with 3D Control.



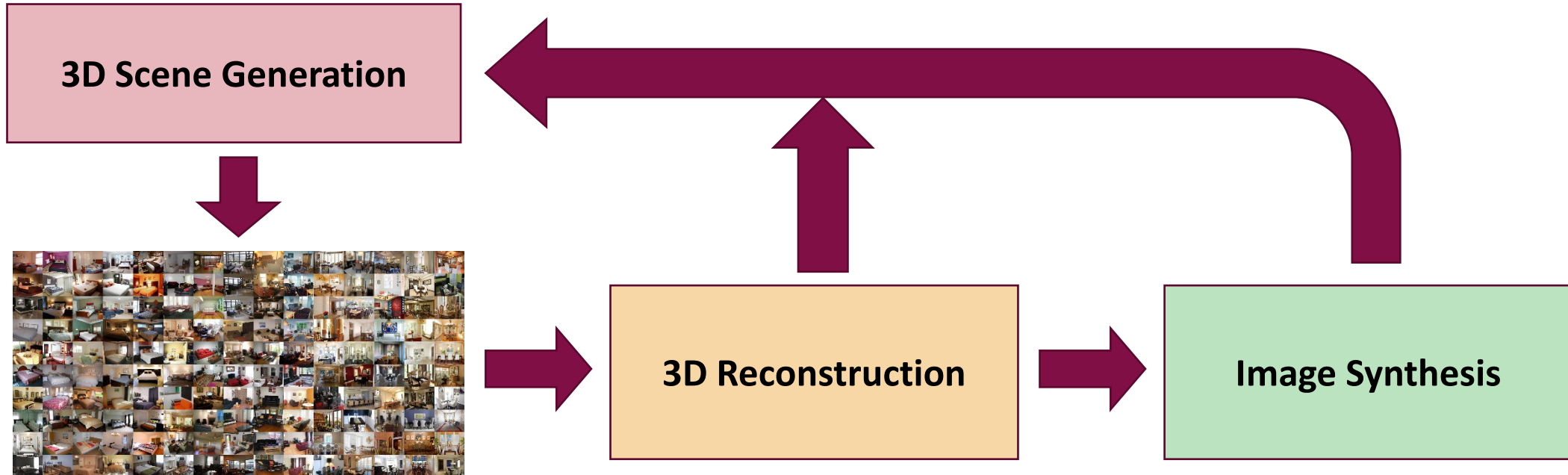
# My Research Interests

- Large-scale 3D Scene Generation



# My Research Interests

- Large-scale 3D Scene **Generation**





# Grading Scheme

- 4x Homeworks: 36%
- Midterm: 20%
- Final Exam: 20%
- Small Projects: 24%
- Possible bonus for extensive in-class participation (up to 5%)
  - I need your interaction and questions. Do not let the communication become one-way! Interrupt at any time.
- Lenient letter grades. The only way to fail this class is to violate the honor code (cheat, plagiarize)
  - Zero Tolerance. Think before you act: One homework or one question in a midterm is unlikely to affect your final letter grade.

# Spring 2026 Tentative Schedule

| Week | Date*                    | Topics                         |
|------|--------------------------|--------------------------------|
| 1    | Jan 20                   |                                |
| 2    | Jan 27                   |                                |
| 3    | Feb 3                    |                                |
| 4    | Feb 10                   |                                |
| 5    | Feb 17                   |                                |
| 6    | Feb 24                   |                                |
| 7    | Mar 3                    |                                |
| 8    | Mar 10<br>(Spring Break) |                                |
| 9    | Mar 17                   |                                |
| 10   | Mar 24                   | <b>Midterm (20%, date TBD)</b> |
| 11   | Mar 31                   |                                |
| 12   | Apr 7                    |                                |
| 13   | Apr 14                   |                                |
| 14   | Apr 21                   |                                |
| 15   | Apr 28                   |                                |
| 16   | Btw May 4-12             | <b>Final Exam (20%)</b>        |

HW1 (Feb 3 – Feb 17)

10%

HW2 (Feb 17 – Mar 3)

10%

HW3 (March 3 – March 24)

8%

HW4 (March 31– Apr 14)

8%

two small projects(Apr 14 – May 14)

24%

HW will be released  
on Thu of that week  
and will be due two  
weeks later, on Thu.

HW plans might evolve

# Waitlisty

- There is typically a 2–3 day delay for the system to reflect my approval on your end after I process your request. Please be patient.
- Currently, the course is at maximum capacity, constrained by our classroom size. So even if I approve your request on my side, you still won't be able to register. Sorry about that.

# Course Team Spring' 26



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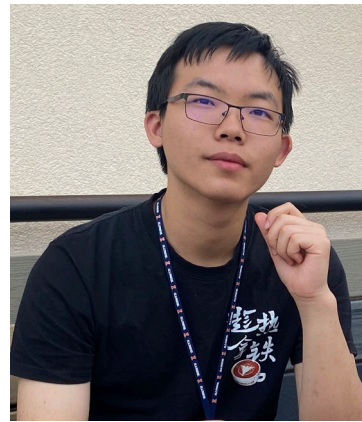
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# How to Speak With The Course Team

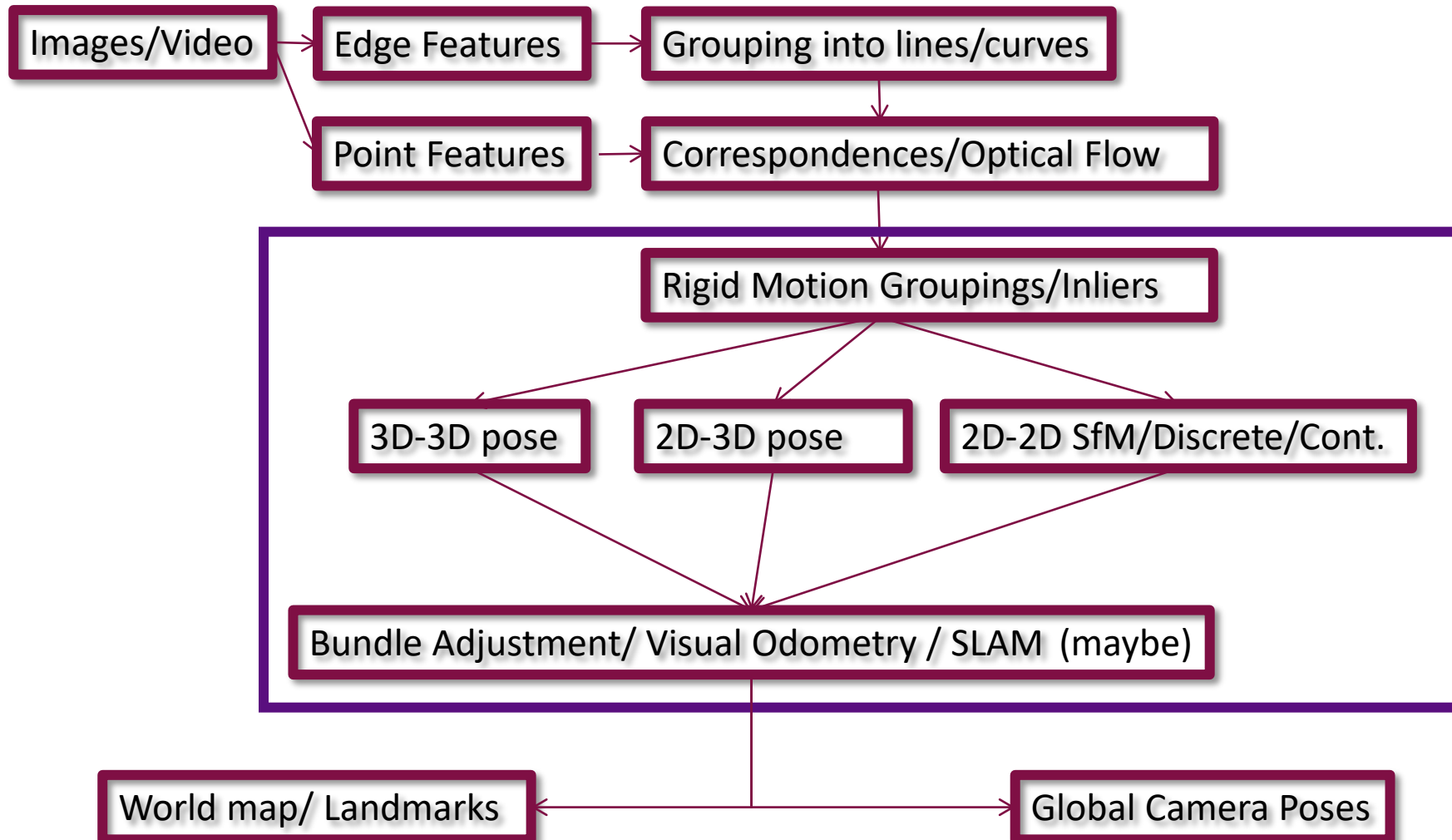
Go to office hours!

- Each of the TAs will have 1 hour of OH each week.
- I will also have 1 hour of OH each week.
- Starting from next week (the week of Jan 26)

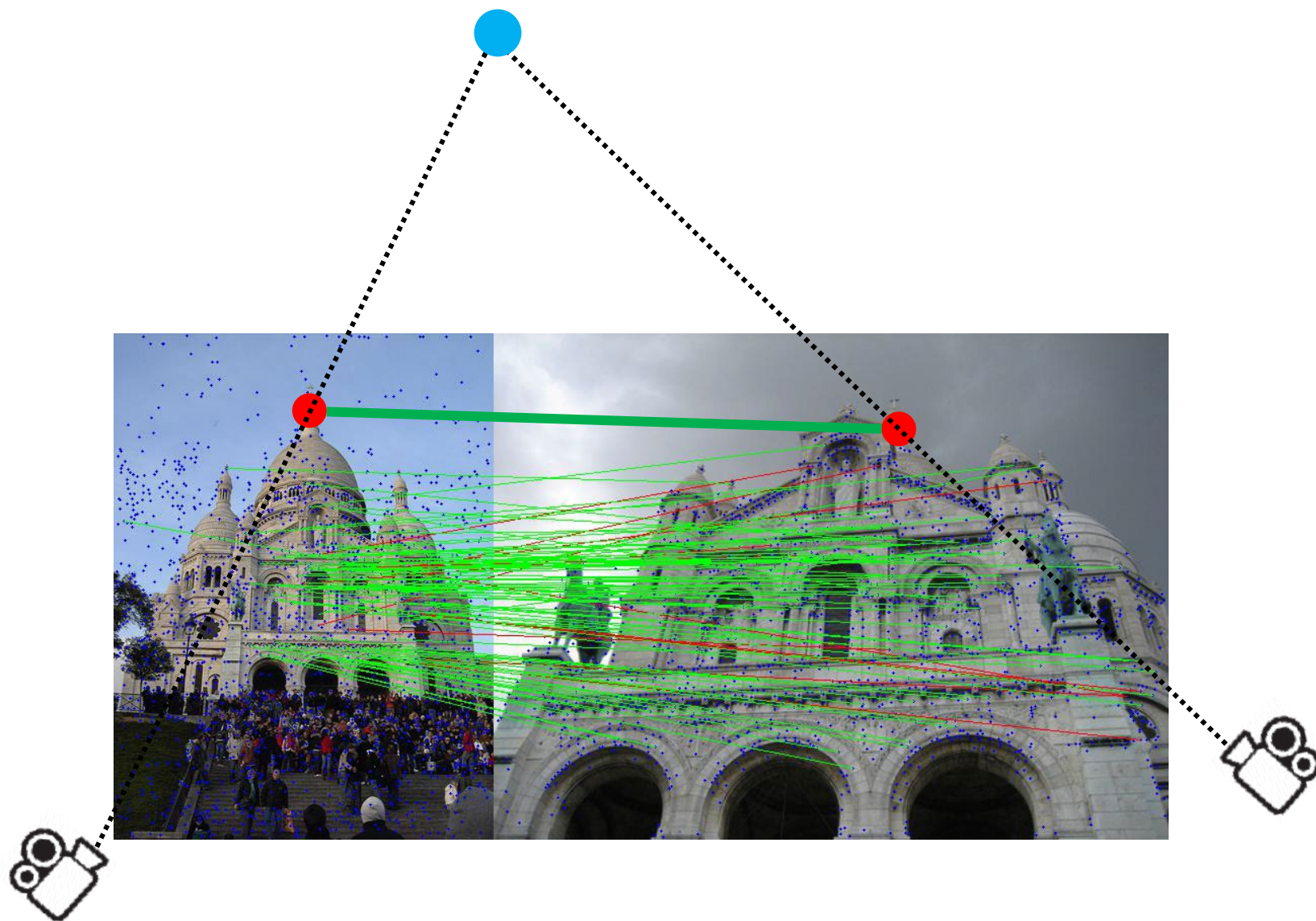


# Lec 1: Perspective Projection

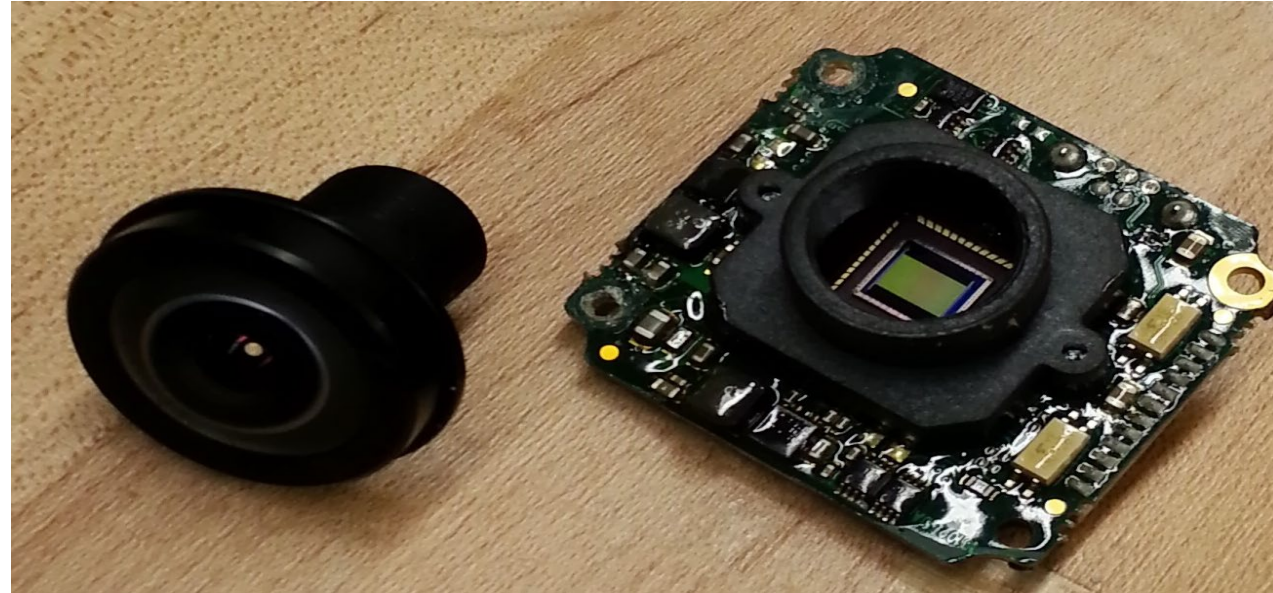
# Geometric Perception Pipeline







# What is a camera?

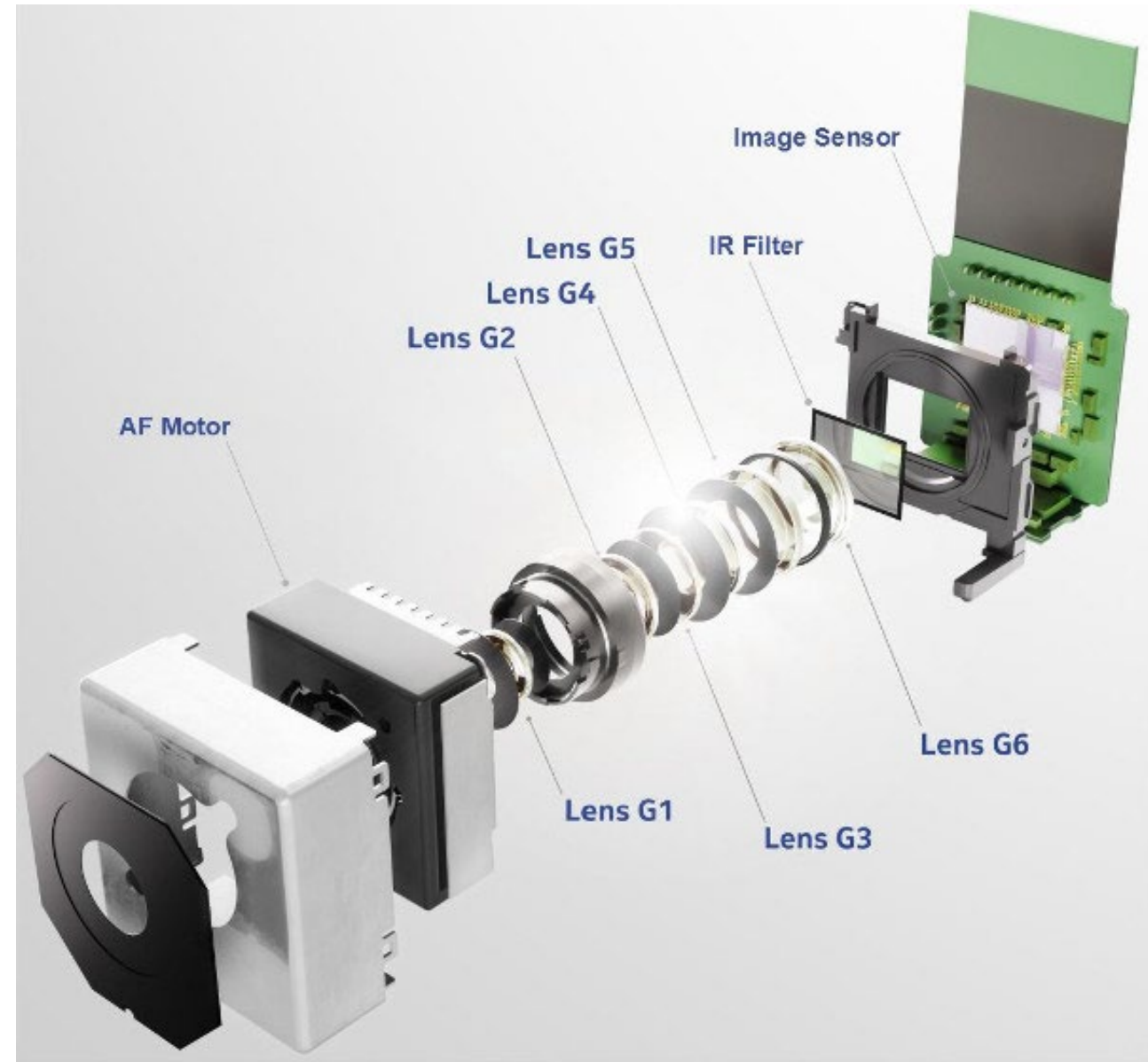


Most basically, an imaging sensor, and a lens

# But real cameras hide many complex details!

- Compound lenses containing many plastic / glass components
- Moving lenses for autofocus
- Lens distortions etc.

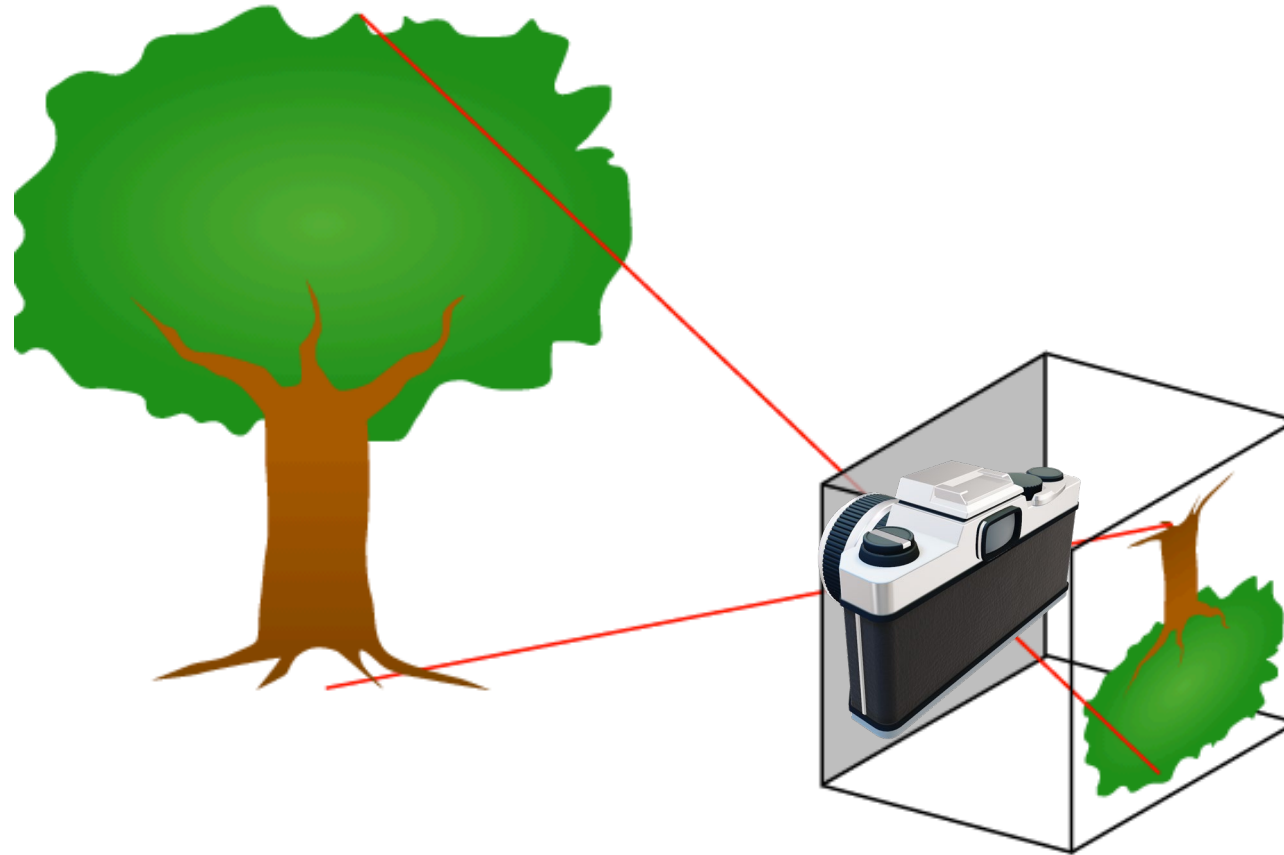
For the most part, we can ignore these complex optics when studying geometric vision, and then deal with them through minor corrections afterwards.



# Introduction to Perspective Projection



# Perspective Projection => Pinhole Camera Model

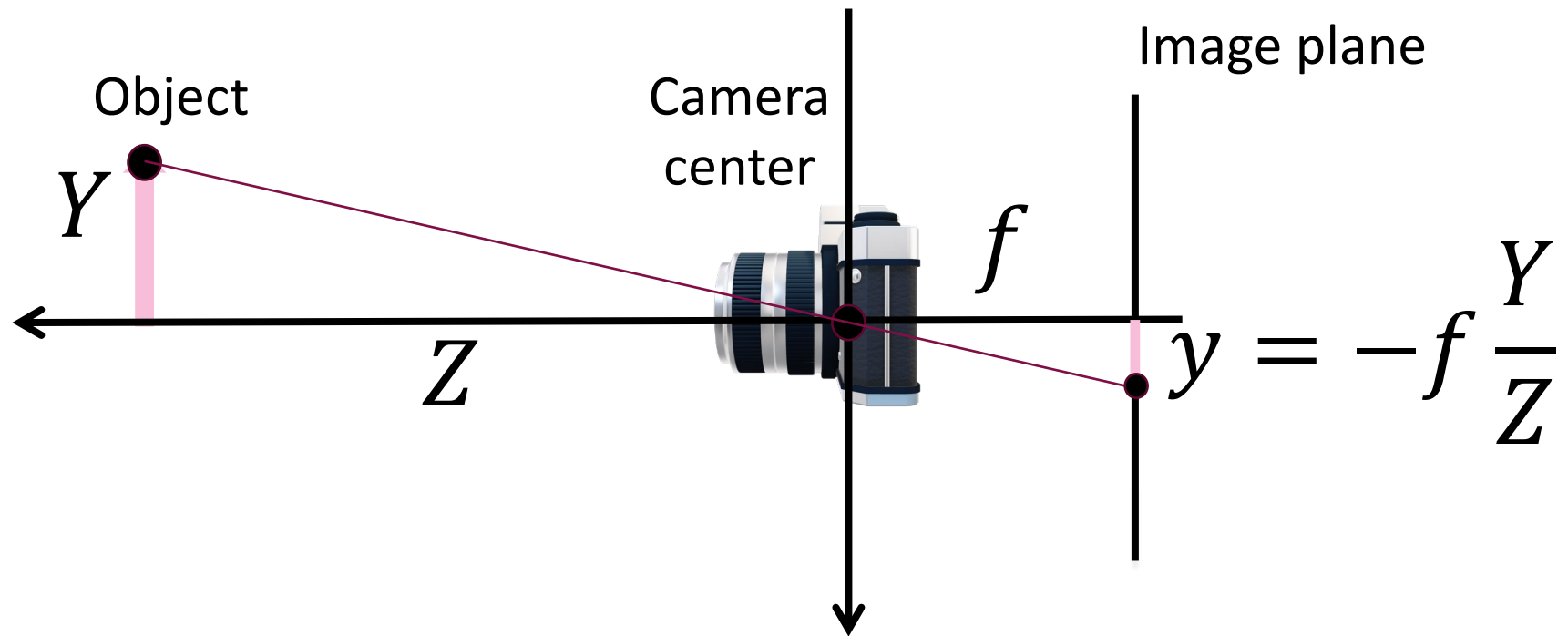


This is a good model for a camera with a lens, as long as the “aperture” is small.



# Pinhole camera model (visualized in 2D)

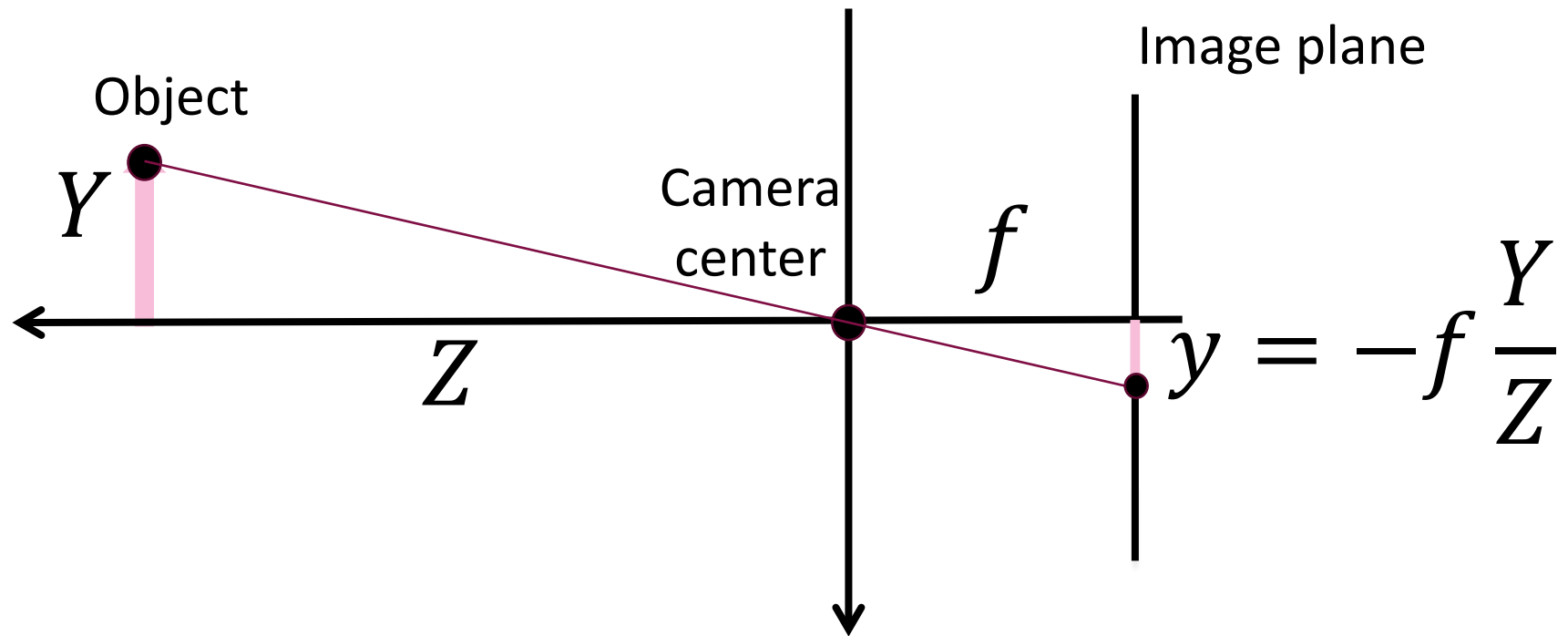
$f$  = distance of camera center from image plane



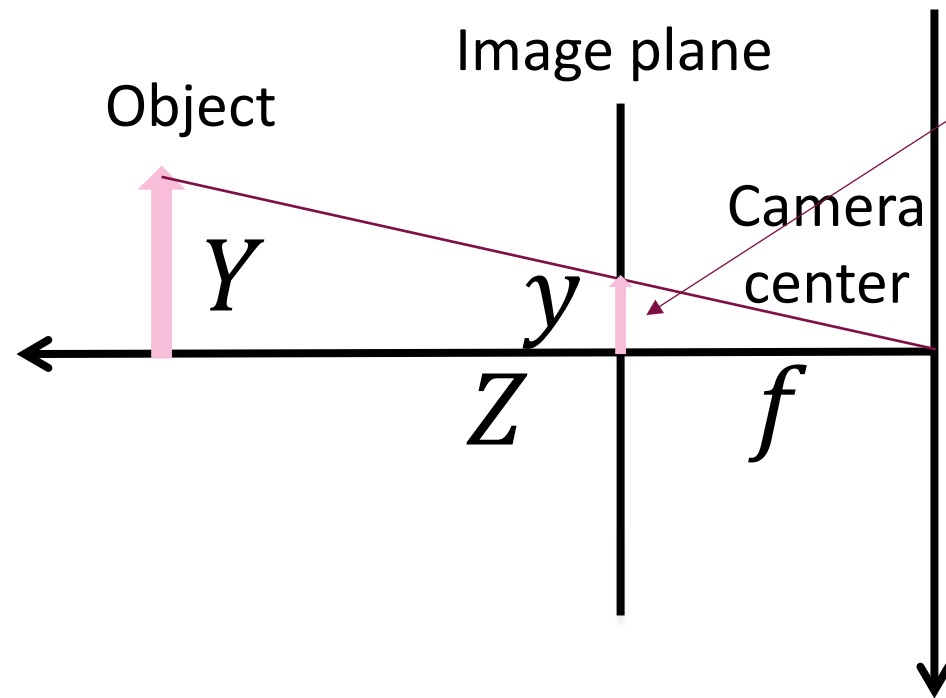
Note how we are expressing all positions like  $Y, Z, y$  in coordinates tied to the camera. We will revisit this choice soon.

# Pinhole camera model (visualized in 2D)

$f$  = distance of camera center from image plane



# Pinhole camera model with the image plane in front

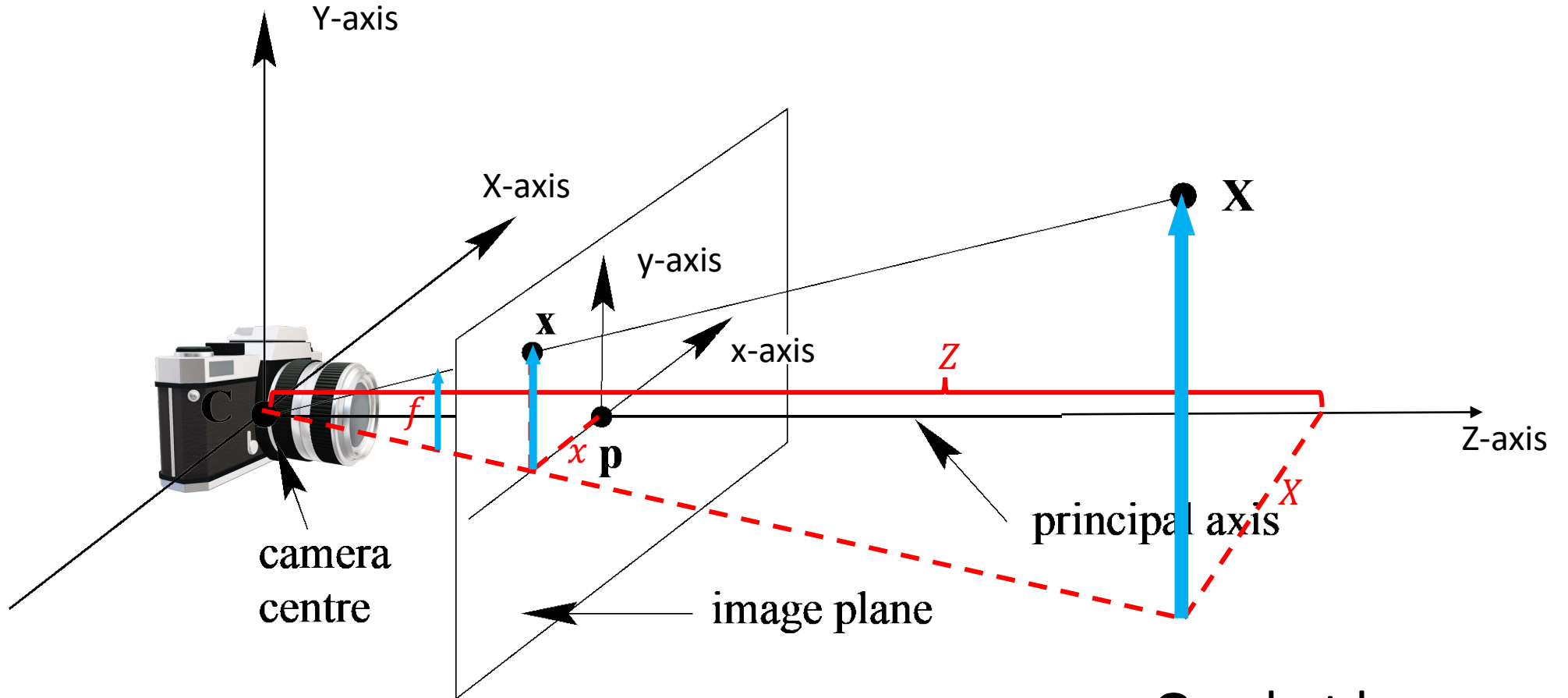


$$y = f \frac{Y}{Z}$$

For convenience move the image plane in front of camera so that objects appear upright.

Can get rid of the pesky negative sign.

# Basic Perspective Projection Equations, now in 3D



$$x = f \frac{X}{Z}, y = f \frac{Y}{Z}$$

**Q:** what happens to sizes of objects as you move the image plane close or far?

# Perspective Projection in Renaissance Art

Albert Durer (1471-1528)

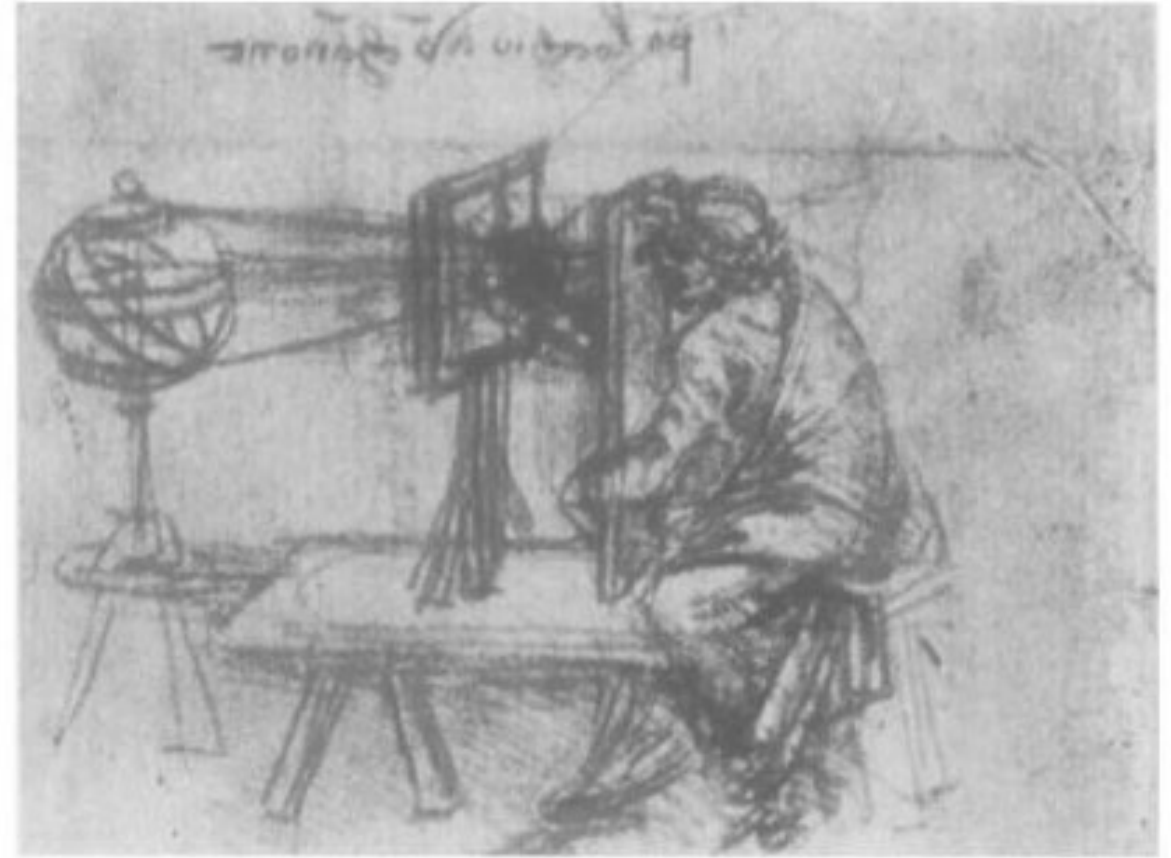
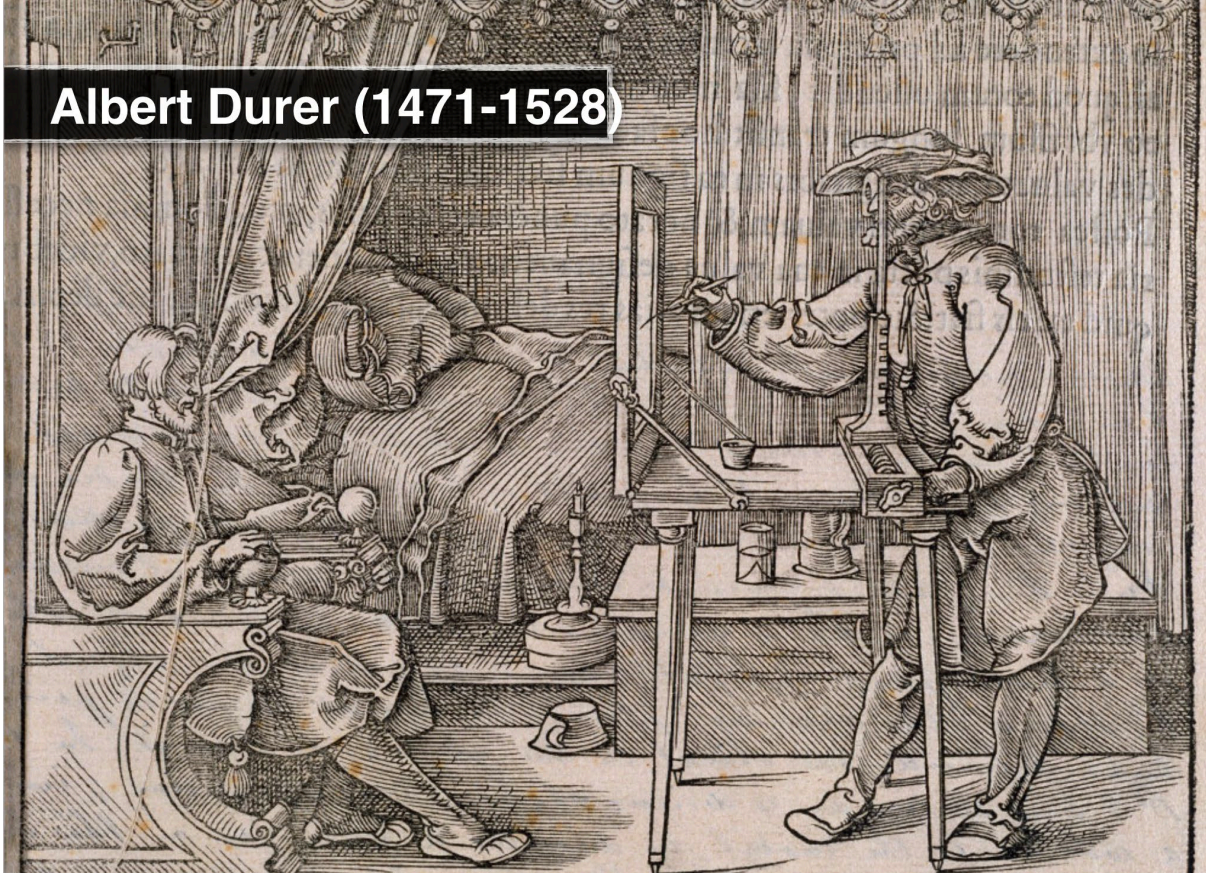


Figure 2.1. Leonardo's technique for making a perspectival drawing of the sphere of the macrocosm (CA 1 ra bis).







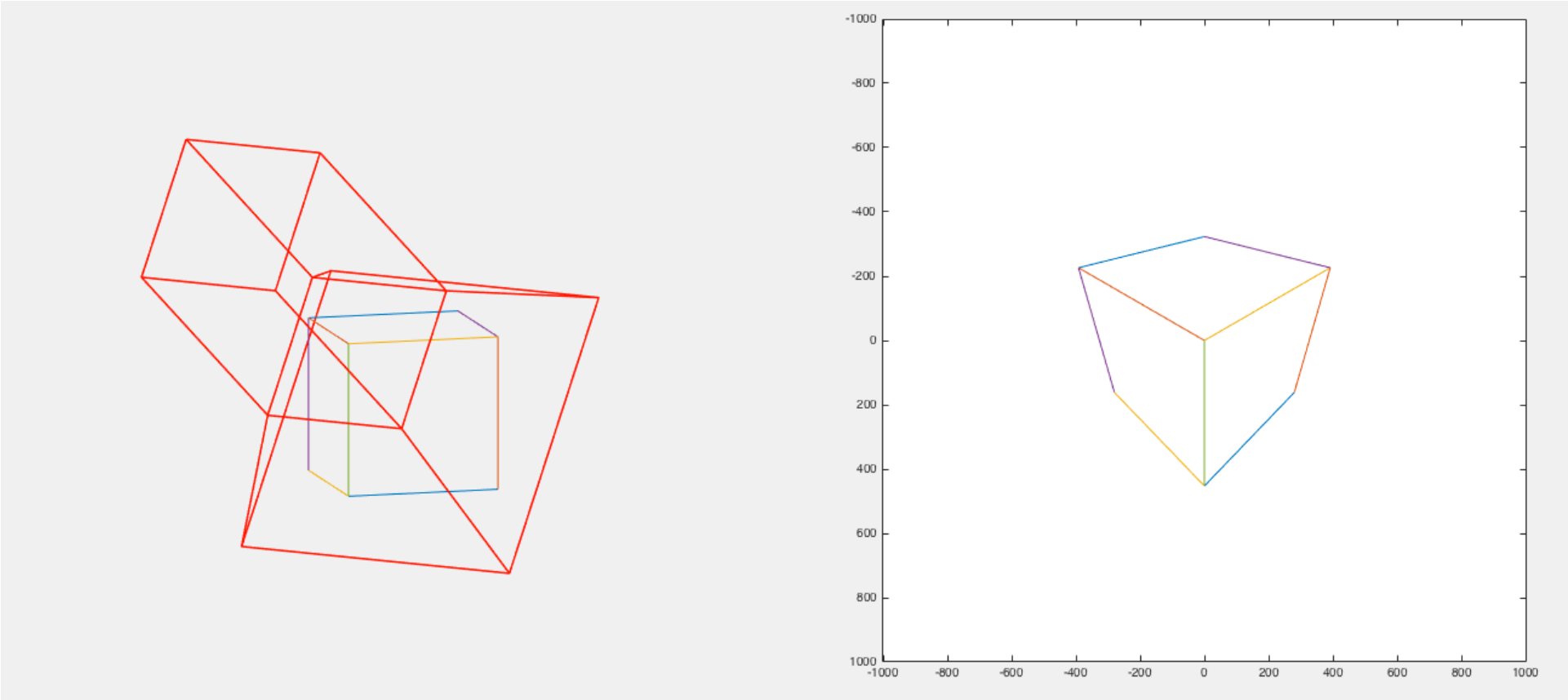
# What properties are preserved by perspective projections?

- Shape?
  - e.g. square, circle, etc.?
- Parallelness / Angles?
- Lengths?
- Ratio of lengths?

**None of the above!**

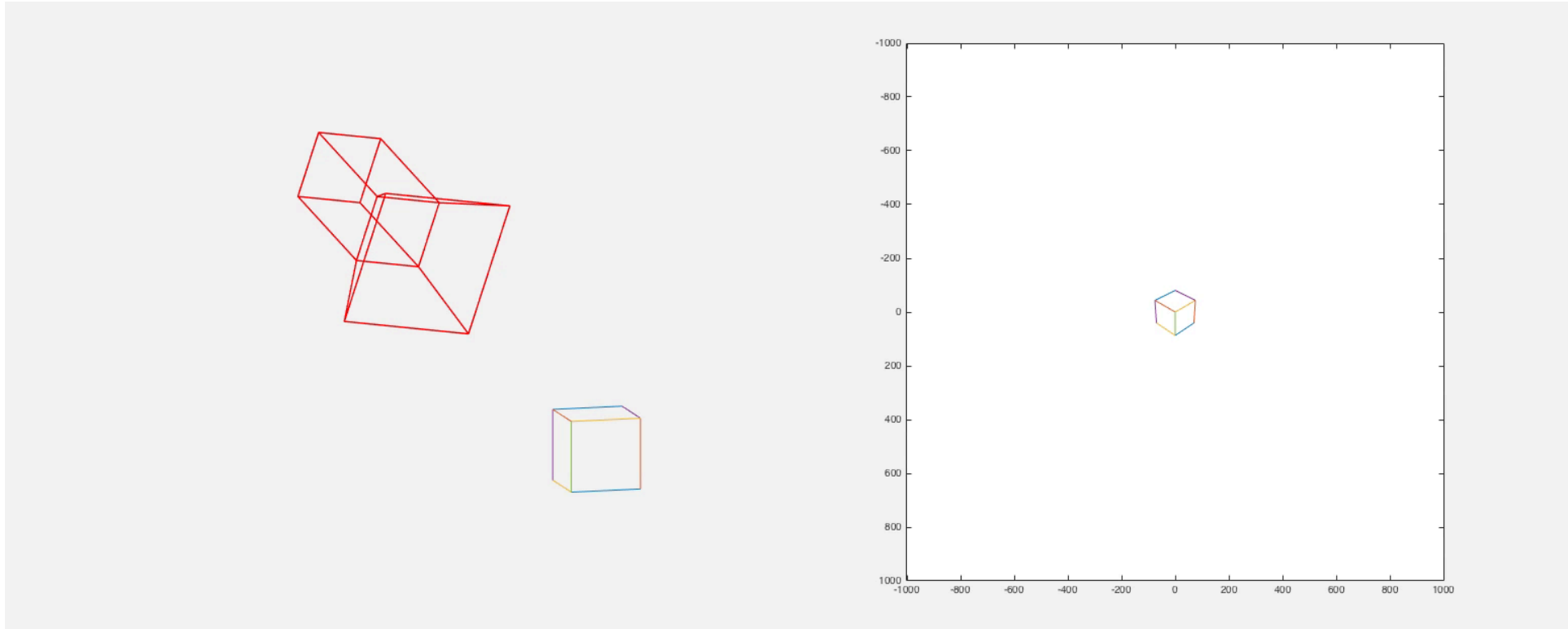
A projection is only required to preserve “straightness”/“collinearity”.

# Perspective effects



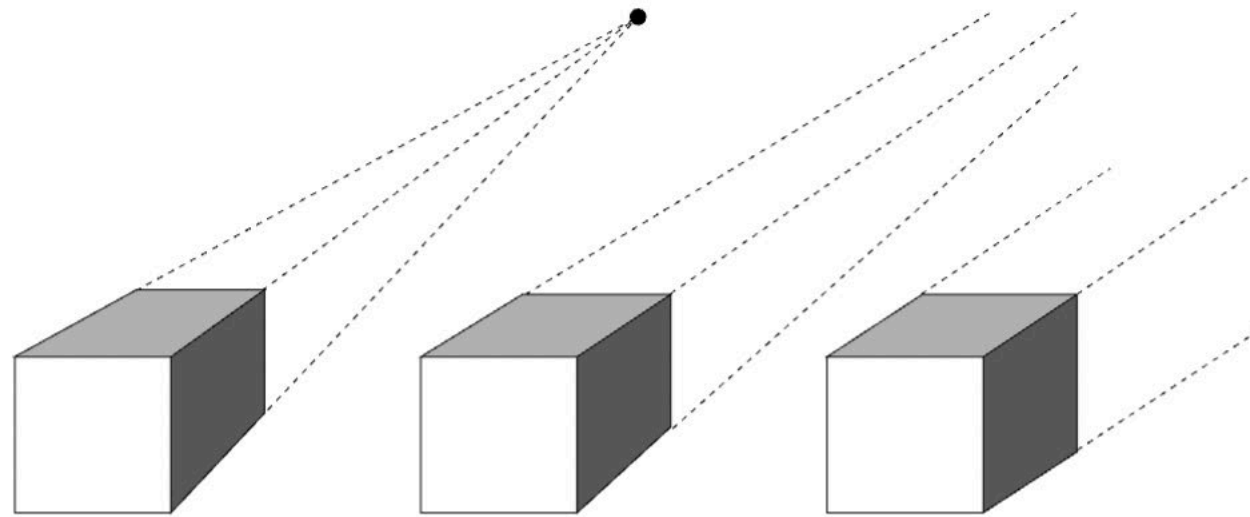
Parallel lines do not remain parallel !

# Perspective effects



Objects decrease in size with distance



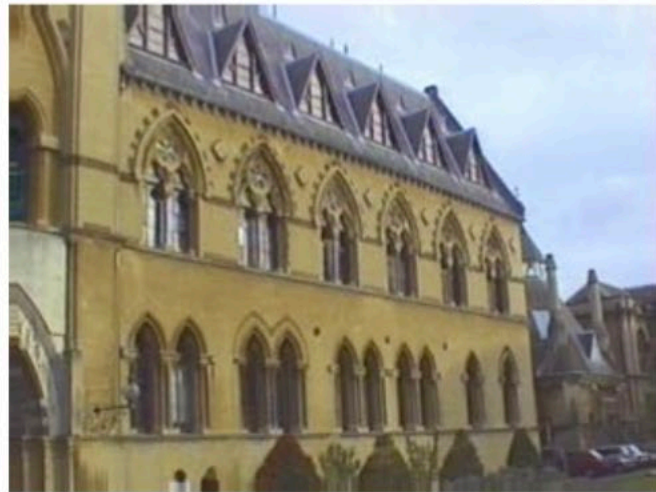


**perspective**

**weak perspective**

————— **increasing focal length** —————→

————— **increasing distance from camera** —————→

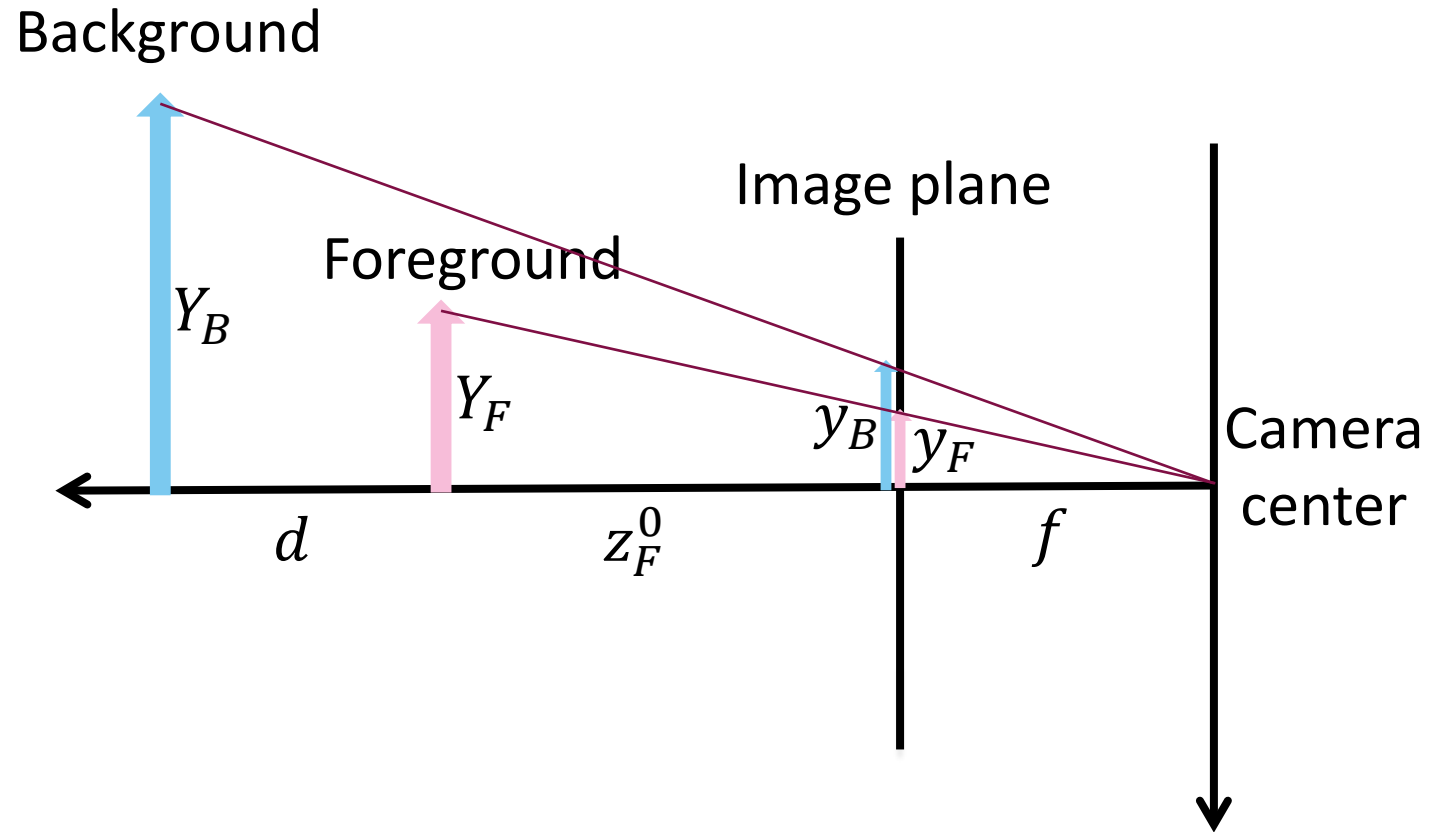


## Dolly Zoom or “Vertigo” Effect: z-motion and Zoom



Goodfellas, Martin Scorsese, 1990

# Dolly Zoom or “Vertigo” Effect: z-motion and Zoom

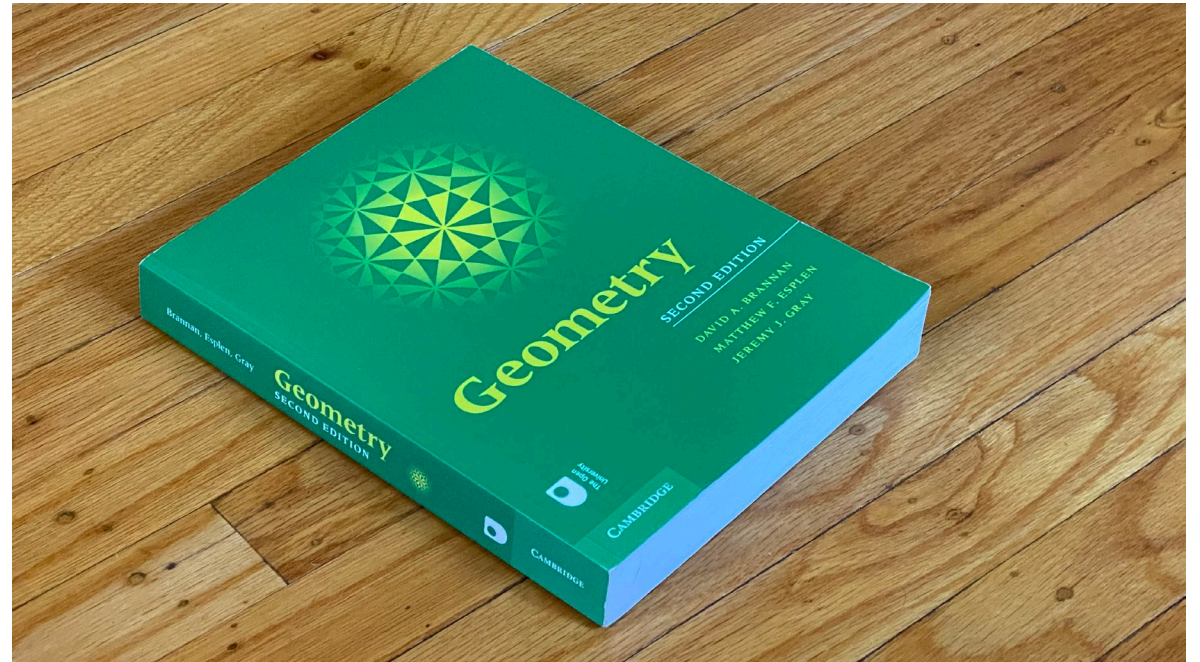
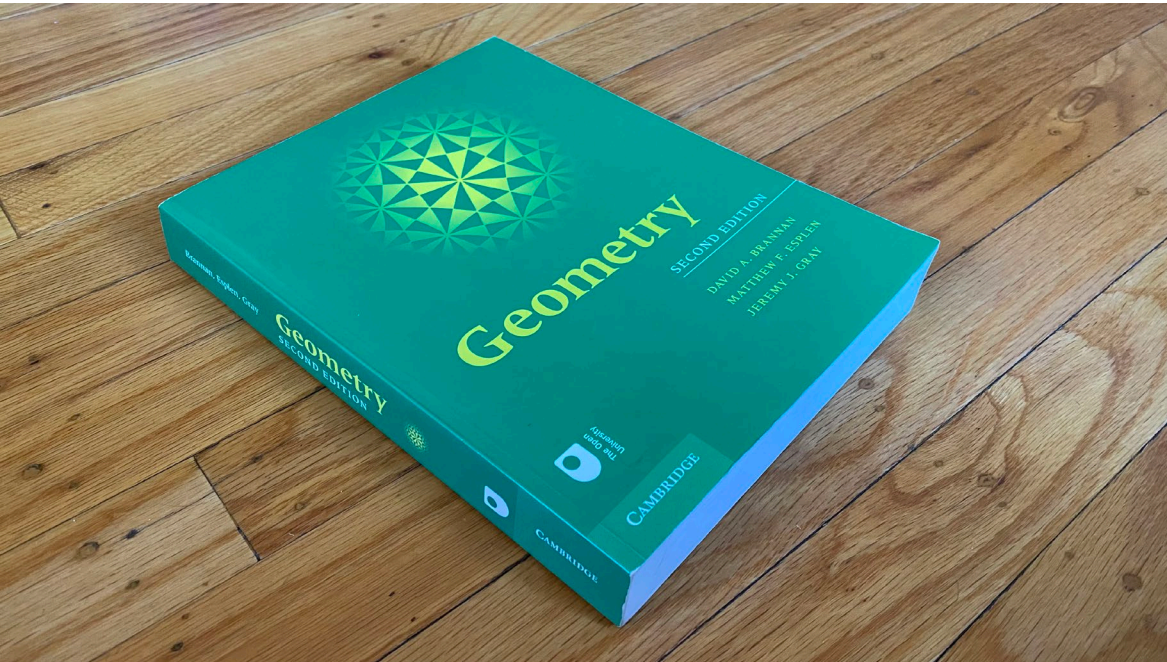


$$y_F = f \frac{Y_F}{Z_F}$$

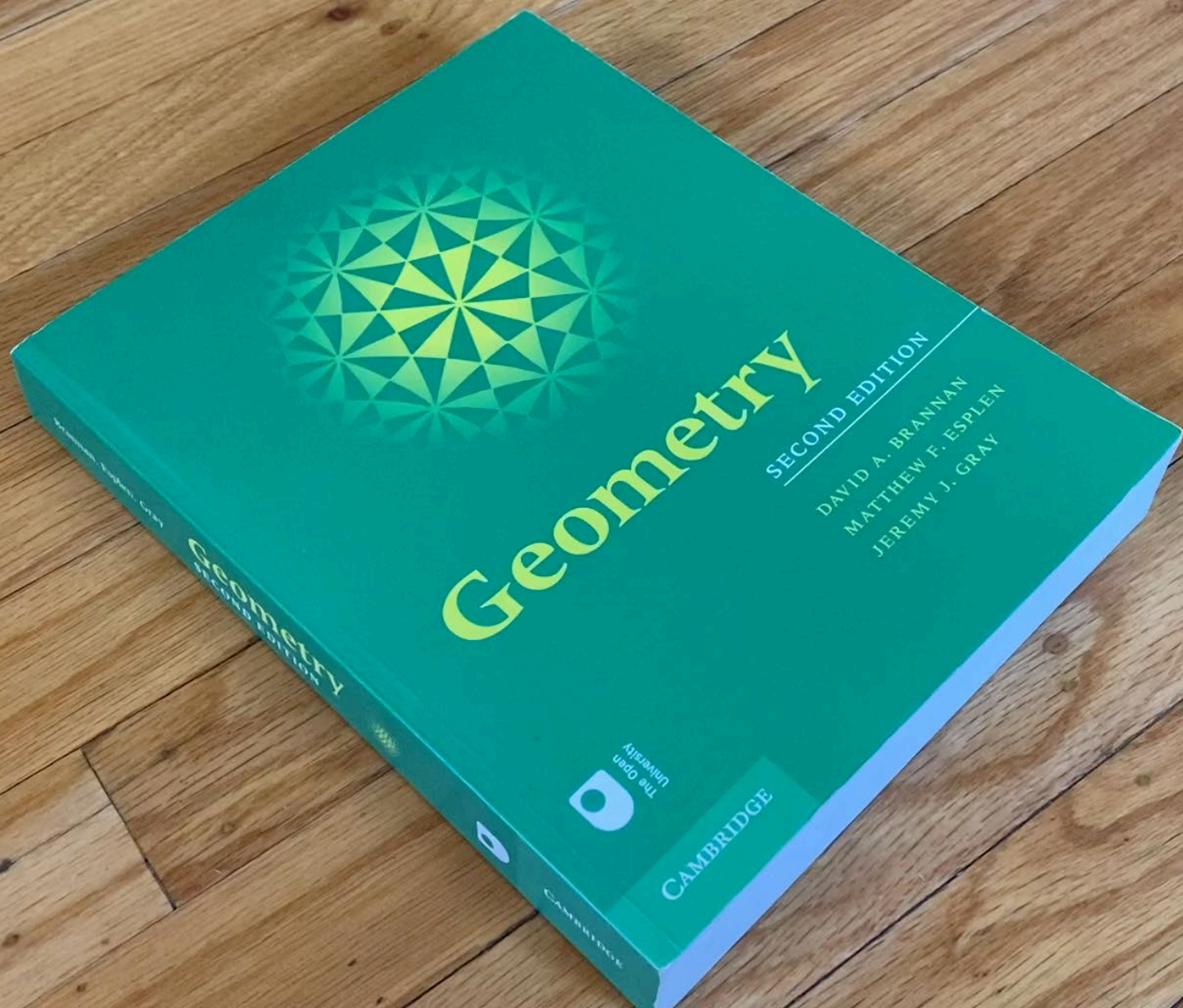
$$y_B = f \frac{Y_B}{Z_B}$$



# Zoom vs distance...







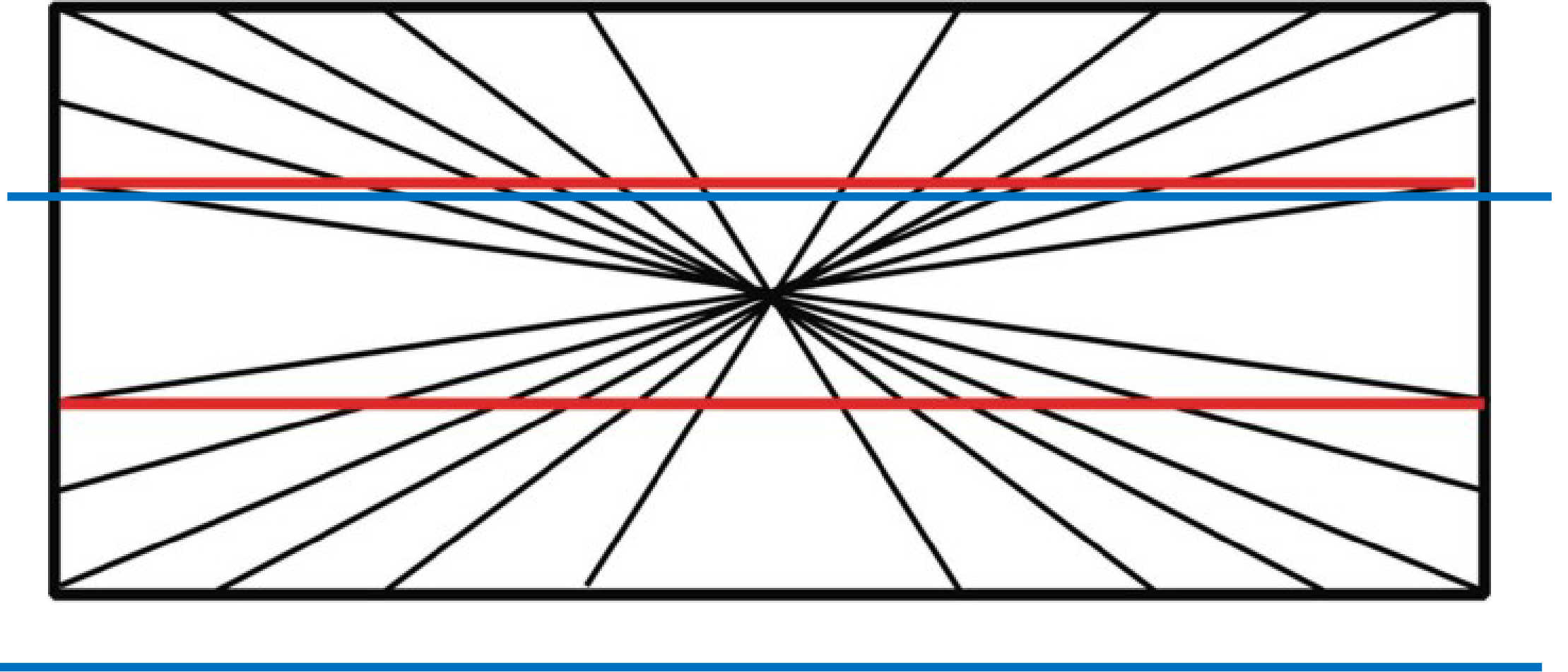


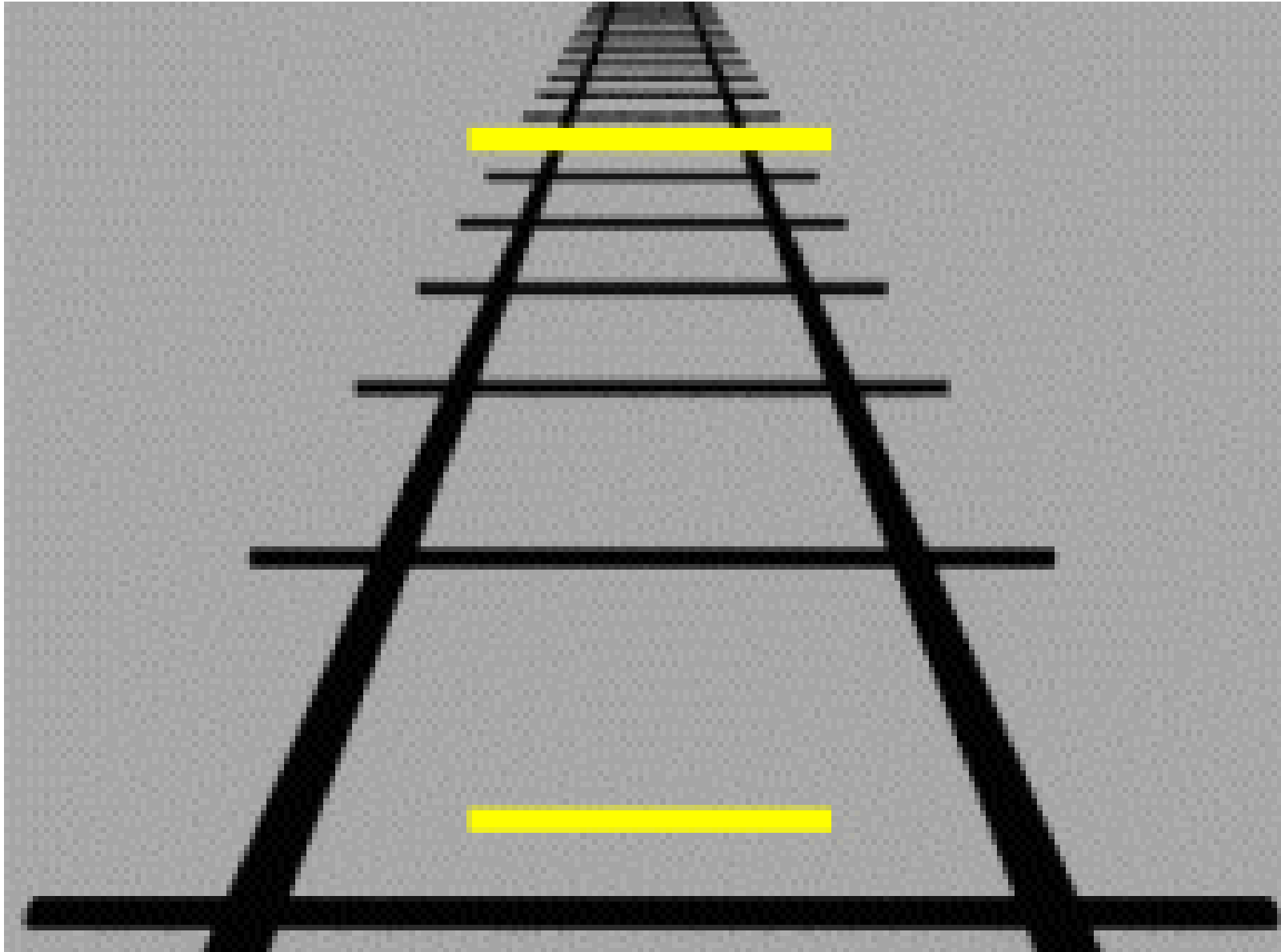


Human visual system is aware of perspective effects!



# Visual Illusion



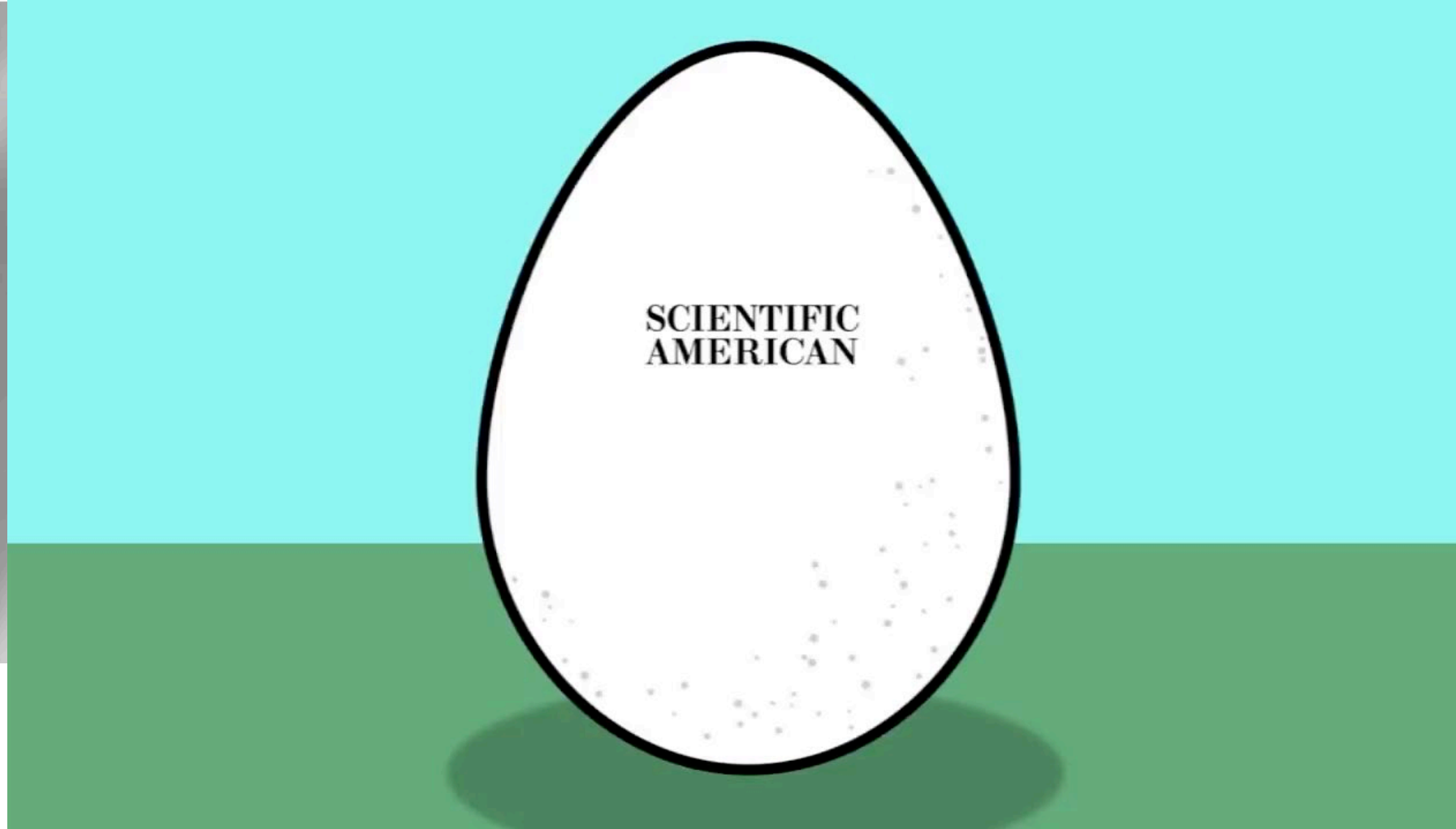
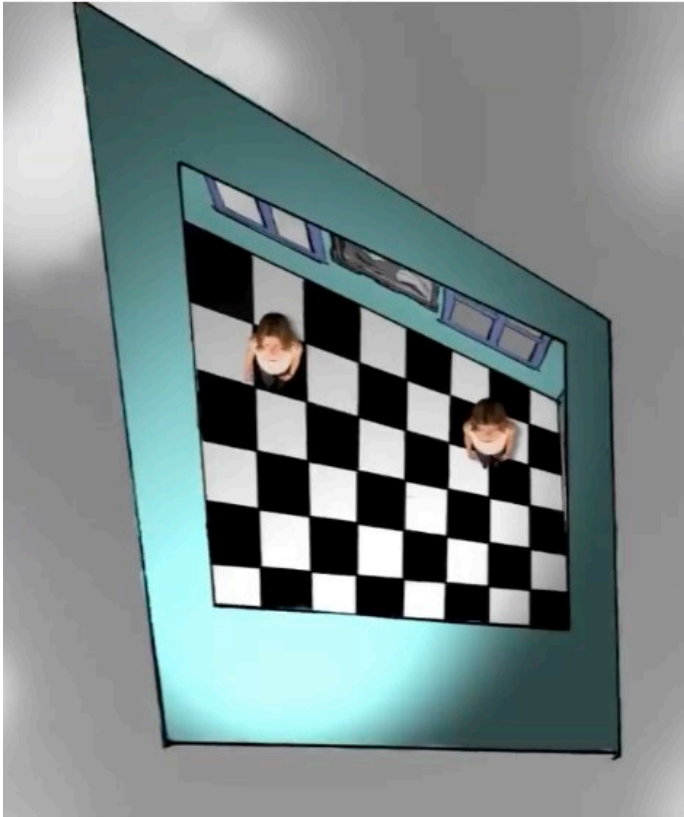




SCIENTIFIC  
AMERICAN



# Ames Room Illusion



<https://www.scientificamerican.com/video/instant-egghead-what-is-the-ames2012-10-09/>

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# Vector Algebra Recap: Dot Products

- $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^T \mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle = \sum_i x_i y_i = ||\mathbf{x}|| ||\mathbf{y}|| \cos \theta_{xy}$ 
  - Output is a scalar
  - Measures angles if applied to unit vectors.
  - Measures projection of one vector onto another:  $||\mathbf{y}|| \cos \theta_{xy}$  is projection of  $\mathbf{y}$  onto the direction of  $\mathbf{x}$
- Orthogonality: Two vectors are orthogonal iff  $\mathbf{x}^T \mathbf{y} = \mathbf{0}$ 
  - The equation of a plane in 3D is  $\boldsymbol{\pi}^T \mathbf{x} = \mathbf{0}$ 
    - This means the vector  $\boldsymbol{\pi}$  is normal to the plane
    - (same reasoning also applies for a 2D line  $\mathbf{l}$ )
- $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$



# Vector Algebra Recap: Matrix Products

- $M\mathbf{x} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}_{2 \times 2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1} = \begin{bmatrix} m_{11}x_1 + m_{12}x_2 \\ m_{21}x_1 + m_{22}x_2 \end{bmatrix}_{2 \times 1}$
- Can be reframed in terms of dot products.
  - $\mathbf{m}_1 = [m_{11}, m_{12}]$
  - $\mathbf{m}_2 = [m_{21}, m_{22}]$
  - Then  $M\mathbf{x} = \begin{bmatrix} \mathbf{m}_1 \cdot \mathbf{x} \\ \mathbf{m}_2 \cdot \mathbf{x} \end{bmatrix}_{2 \times 1}$

